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भारतीय मानक मसौदा
शल्य चिकित्सा के लिए द्रव-चालित मेज - विशिष्टि
[IS 5291 का पहला पुनरीक्षण]

Draft Indian Standard
Hydraulic Operation Table - Specification
[First Revision of IS 5291]

ICS 11.140

Hospital Equipment and Surgical Disposable
Products Sectional Committee, MHD 12

Last date for comments:
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FOREWORD

(Formal clause will be added later)

This standard was originally published in 1969. The first revision of this standard has been brought out to align it with recent developments in the material requirements, addition of an operating manual and to bring the standard in line with the latest style and format of referred standards.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2: 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This standard specifies the construction and performance requirements for manually controlled hydraulic operation tables.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

<i>IS No.</i>	<i>Title</i>
IS 1300: 1994	Phenolic moulding materials - Specification (<i>third revision</i>)
IS 6911: 2017	Stainless steel plate, sheet and strip - Specification (<i>second revision</i>)
IS 1068: 2025 ISO 1456: 2009	Metallic and other inorganic coatings - Electrodeposited coatings of nickel, nickel plus chromium, copper plus nickel and of copper plus nickel plus chromium (<i>fourth revision</i>)
IS 17523: 2021	Castors and Wheels - Requirements for Castors for Furniture

3 TERMINOLOGY

3.1 Operation Table - The surgical operation table is fitted with a hydraulic pump to raise and lower the table top with a head-end control for positioning of the table top during surgical operations.

3.2 Jerk - The sudden change in speed or quick movement.

4 MATERIAL REQUIREMENTS

4.1 The materials used in the manufacture of the operation table shall be corrosion-resistant or protective finished.

4.2 The raising column and the gearing system shall be made from materials to resist wear during normal manipulation.

4.3 The handles provided shall be made from ebonite or plastics as per Grades 1 or 2 of IS 1300.

4.4 The stainless steel used shall conform to designation X04Cr18Ni10 or X07Cr17Ni12Mo2 in accordance with IS 6911.

4.5 The operation table shall have an antistatic rubber tread castor as per IS 17523 or cotton impregnated Bakelite wheels. The operation table shall have a static charge transmission mechanism to the earth.

5 SHAPE AND DIMENSIONS

5.1 Table Top

The minimum overall length of the table top shall be 180 cm with the head section and 195 cm without the head section. The overall width of the table top excluding side railings shall be 50 to 55 cm.

5.2 Height

The height of the table top without mattress shall be 75 ± 7.5 cm and in the extreme lifted position 115 ± 5 cm subject to limitation of the hydraulic lift as specified in **6.4**.

6 CONSTRUCTION REQUIREMENTS

6.1 Design

A typical design for the operation table shall be as per Fig. 1.

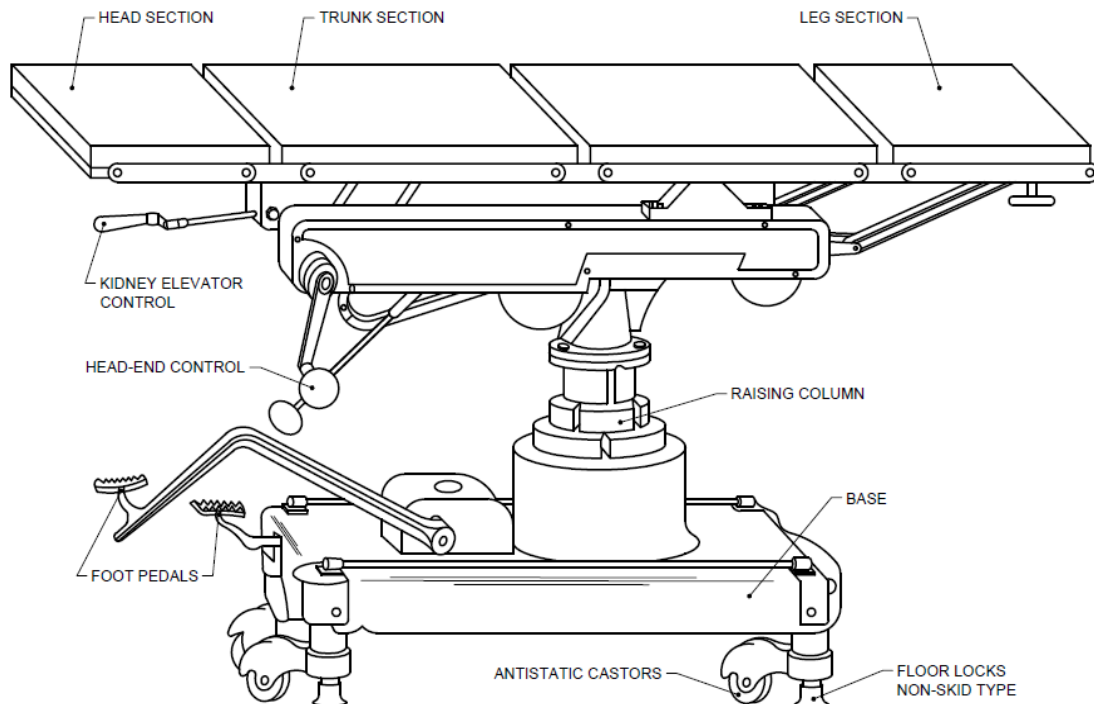


Fig. 1 HYDRAULIC MAJOR OPERATION TABLES

6.2 Base

The base shall be solid and mounted on castors to provide stability to the operation table. The base shall have non-skid compensating floor locks actuated by a foot pedal. The base may also have foot rails for providing a comfortable footrest for the surgical team.

6.3 Table Top

The table top shall have a minimum of four sections namely, head, trunk (two sections) and leg sections. The head section shall be detachable. The detachable feature for the leg section depends on manufacturer and purchaser requirements. The entire table top shall have a jerk-free vertical movement.

The multiple cylinder hydraulic systems may have sudden change in speed (jerk) as a result of cylinder diameter differences which shall not be in excess of ' $n - 1$ ', where ' n ' is the number of cylinders. The minor operation table shall also satisfy the following requirements:

- The entire table top shall be capable of being tilted to $45^\circ \pm 3^\circ$ in Trendelenburg and reverse Trendelenburg positions.
- The table shall have arrangements for a lateral tilt of $20^\circ \pm 2^\circ$ in either direction.
- The table shall allow angulation in flex (90°) and reflex (220°) positions in one or several cranking operations (see Fig. 2).
- The table top may be provided with a perineal cut-out for the drainage tray.
- The table top shall have a separate kidney elevator if there is no specific arrangement to obtain kidney position which is capable of raising at 150 mm from the table top and shall have a head-end control for manipulation.
- The entire table top shall have a sweep of 360° in the horizontal plane as required by the purchaser.

6.3.1 Head and Leg Sections

The head section shall be able to raise 40° to 45° and in its reverse position 120° from the trunk portion. The head section shall be so designed to be attached on the leg section side in the sitting position. The leg section shall be capable of being raised or lowered to 90° from the trunk so as to enable the table to be used as a chair.

The locking mechanism shall be provided at stages of every 15° angle in the sitting position to prevent wobbling of the foot section. The head section and the table top shall have controls capable of positioning during operation and which shall not interfere with the surgical team.

The controls are situated on the head-end side and the various controls may be independent or at one single control point actuated through a gear mechanism with an indicator dial shall be controlled by a common crank or wheel with folding handles.

6.4 Hydraulic Lift

The table shall have a hydraulic pump encased in the base for controlling the lift of the table. The hydraulic pump shall be operated by a pedal. The table lift shall be between 30 to 50 cm.

7 ACCESSORIES

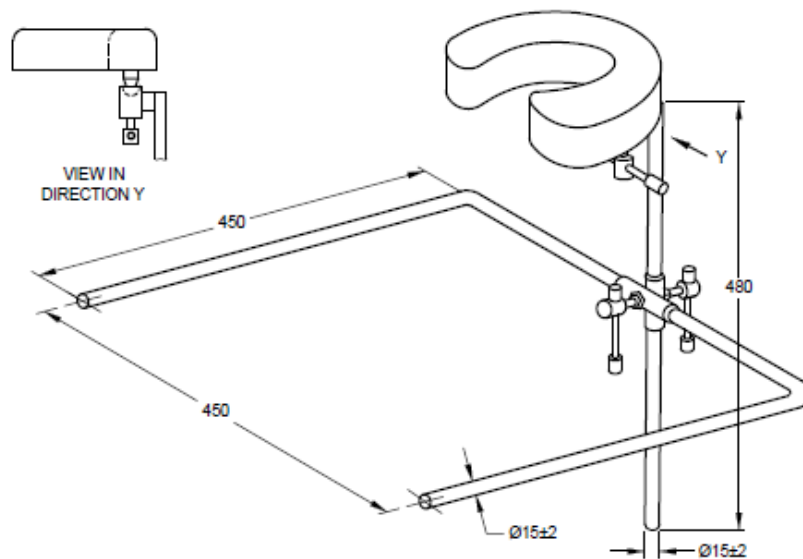
7.1 The Table 1 shall be provided with the following standard accessories:

Table 1 List of Accessories

Sr. No.	Accessories	Count in (No.)
1	Foot extension	1
2	Shoulder supports	2
3	Lateral supports or side supports	2
4	Anaesthetic screen	1
5	Kidney elevator	1
6	Arms support	1
7	Pelvic holder strap	1
8	Leg strap	2
9	Wristlets	2
10	Pair of knee crutches/lithotomy leg holders	2
11	Intravenous arm board	1

7.2 The following arrangements shall be optional:

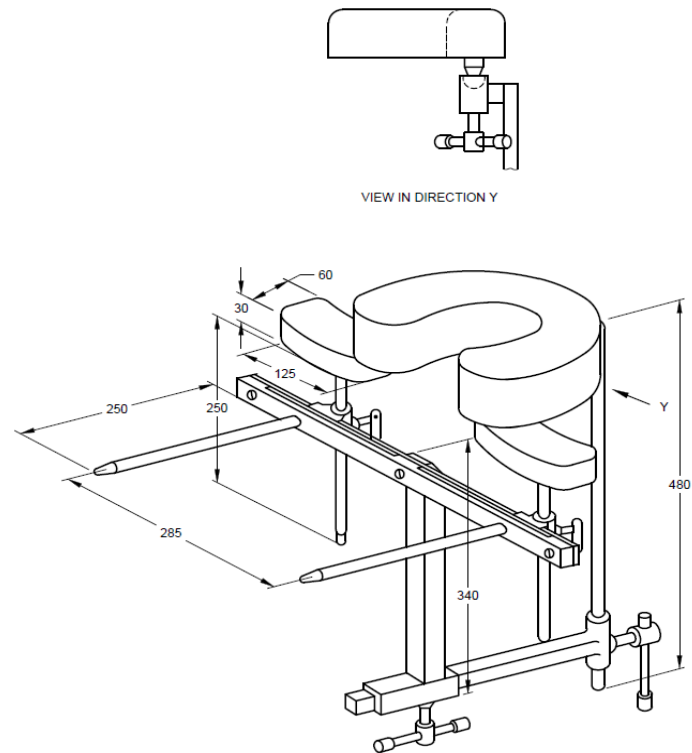
- a) Neck elevator or shoulder bridge,
- b) Permeable table top for radiography with cassette tray and necessary guide,
- c) Lateral cassette holder,
- d) Stainless steel transurethral tray with wire mesh screen,
- e) Instruments tray, and
- f) Neuro-surgical attachments as indicated in Fig. 2 to Fig. 4.



All dimensions in millimetres

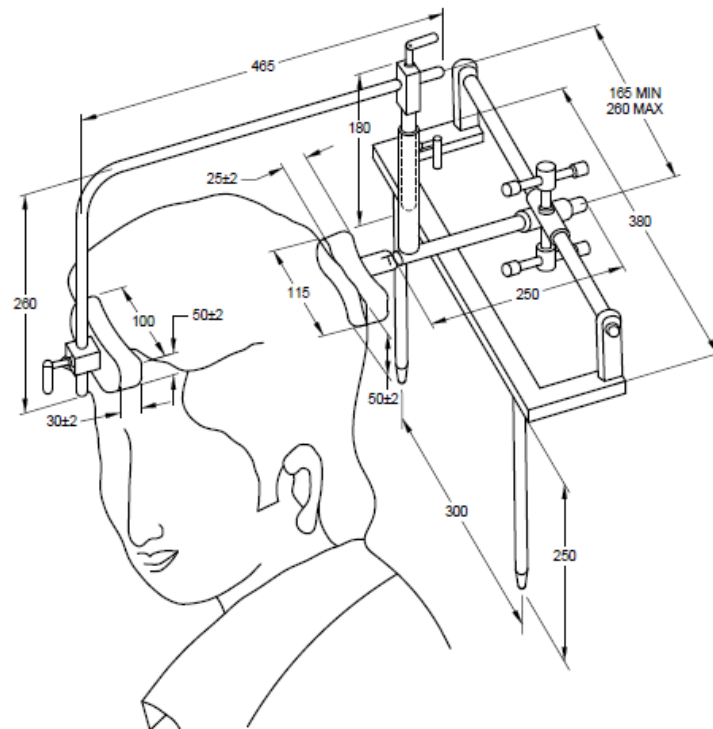
Tolerance on all dimensions ± 5 mm except where specified

Fig. 2 NEUROSURGICAL ATTACHMENT FOR SITTING UP POSITION



All dimensions in millimetres

Fig. 3 NEUROSURGICAL ATTACHMENT FOR PRONE POSITION HEADREST



All dimensions in millimetres

Tolerance on all dimensions ± 5 mm except where specified

Fig. 4 NEUROSURGICAL ATTACHMENT FOR TRIGEMINAL HEADREST

8 PERFORMANCE REQUIREMENTS

8.1 Hydraulic Lift

The hydraulic lift shall give a vertical jerk-free and smooth movement to the system as per 6.3. When the hydraulic pressure is released, the table top shall take travel time of 15 seconds to 45 seconds from the maximum height to the bottom most position. The downward movement shall be smooth and jerk-free as per 6.3. When the table top is held in any position by hydraulic action alone and loaded with 150 kg in the trunk portion the height shall not reduce more than 3 mm in a period of four hours.

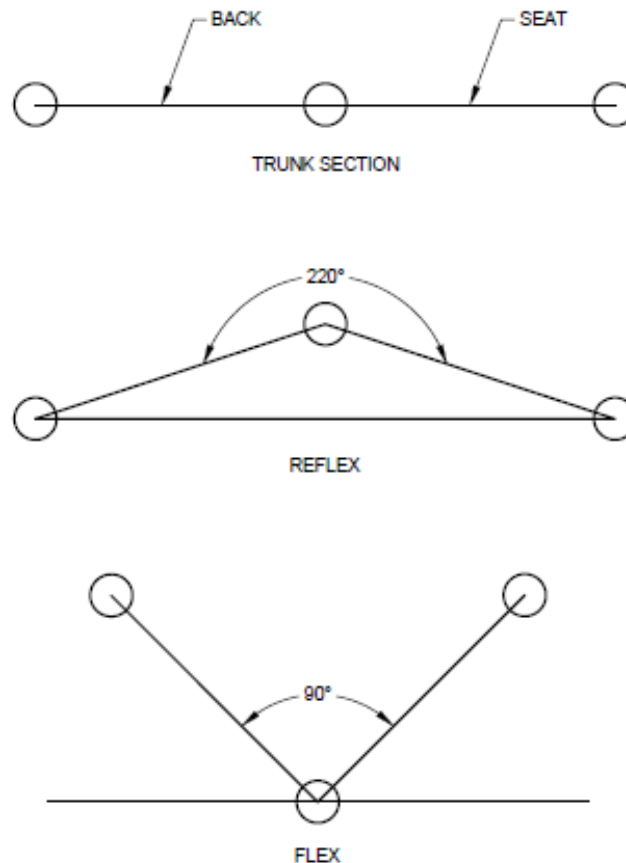


Fig. 5 REFLEX AND FLEX POSITIONS

8.2 Base

The base shall be stable on the castors with non-skid floor locks. The table shall be applied with 12 kg load to move the table on a smooth cemented floor, once the table shall come in motion the 7 kg load shall be sufficient for the motion. The floor locks shall give rigidity to the table and the table shall not move in any direction when a force of 50 kg is applied to the base in any direction.

8.3 Table Top

The table top shall satisfy the requirements given below in any position obtained by the hydraulic lift:

- The extreme edge of the head section shall be applied with a vertical load of 50 kg, when the load is applied the leg section shall not lift by more than 10 mm.

- b) The extreme end of the head section (with the head section drawn in) shall be applied with a load of 5 kg which is dropped from a height of 150 mm, when the load is applied the head section shall not fold in.
- c) The head section shall be pulled along its plane with a load of 25 kg, when the load is applied the head section shall not come off when locked.
- d) There shall be no wobble under normal working conditions when the accessories are clamped.

9 WORKMANSHIP AND FINISH

9.1 The operation table components shall be arranged accurately to give smooth operation and additional performance requirements provided in 8. The operation table surfaces shall be well finished and free from surface defects, such as pinholes, blisters, blow-holes and similar other defects. The edges shall be rounded off and the four sections comprising the table top in their normal position shall be in one plane when tested with a spirit level. The sweeps of the head and leg sections shall be true. The welded joints shall have welds fully penetrating and well finished. The operation table metal surfaces shall be non-corrosive (aluminium and stainless steel) with matt finish.

9.2 The corrosive surface like steel castings for base, cylinder casings, foot pedals shall be painted to shades such as sea green or low reflective shades agreed to between the manufacturer and the purchaser. Initially the surface which is painted shall be degreased, rust-proofed by phosphating and then suitably protected by an anti-corrosive primer with brushing or spraying and then finished by spraying in stove enamel or air-drying enamel of the specified shade. During the procedure each stage shall be separately stoved or air-dried.

9.3 The resulting finish shall be hard and shall not readily chip or flake. The mild steel and brass parts shall be plated chromium over nickel according to **Service Condition 2** of IS 1068.

10 MARKING

10.1 The marking which is used by the manufacturer shall be legibly and indelibly marked with the Identification mark of the manufacturer.

10.2 Each operation table shall have a name plate fixed to it showing the following:

- a) Range of hydraulic lift;
- b) Any special instructions for safe handling of the table.

10.3 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

11 OPERATING MANUAL

11.1 The operation table shall be accompanied by instructions manual giving the following information:

- a) Brief instructions for installation and maintenance (including spare parts), and
- b) Instructions for correct operation of the table.

12 PACKAGING

The operation table shall be packed as agreed to between the purchaser and the manufacturer. Each individual package shall contain only one unit and may be marked with the name of the item, and identification of the source of manufacture and batch number and any other marking if so desired.